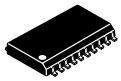
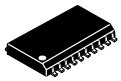
Octal D-Type Flip-Flops with Clear

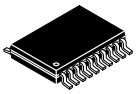
MM74HC273



SOIC−20 WB
CASE 751D−05



SOIC−20, 300 mils
CASE 751BJ−01



TSSOP−20 WB
CASE 948E

General Description

The MM74HC273 edge triggered flip−flops utilize advanced silicon−gate CMOS technology to implement D−type flip−flops. They possess high noise immunity, low power, and speeds comparable to low power Schottky TTL circuits. This device contains 8 master−slave flip−flops with a common clock and common clear. Data on the D input having the specified setup and hold times is transferred to the Q output on the LOW−to−HIGH transition of the CLOCK input. The CLEAR input when LOW, sets all outputs to a low state.

Each output can drive 10 low power Schottky TTL equivalent loads. The MM74HC273 is functionally as well as pin compatible to the 74LS273. All inputs are protected from damage due to static discharge by diodes to V_{CC} and ground.

Features

- Typical Propagation Delay: 18 ns
- Wide Operating Voltage Range
- Low Input Current: 1 μA Maximum
- Low Quiescent Current: 160 μA (74 Series)
- Output Drive: 10 LS−TTL Loads
- This is a Pb−Free Device

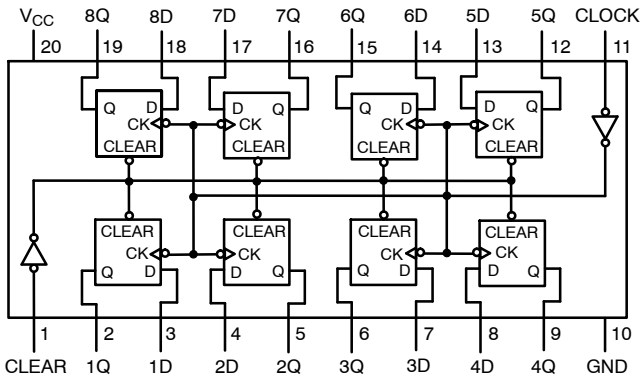
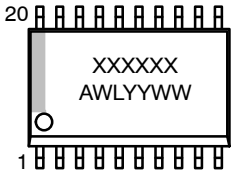


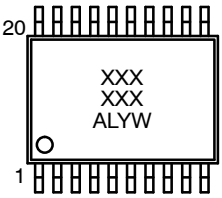
Figure 1. Connection Diagram

MARKING DIAGRAMS

SOIC−20 WB



TSSOP−20 WB



XXXXXX	= Specific Device Code
A	= Assembly Location
WL, L	= Wafer Lot Number
Y	= Year
WW, YW	= Work Week

ORDERING INFORMATION

See detailed ordering and shipping information on page 5 of this data sheet.

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TRUTH TABLE (Each Flip−Flop)

Inputs			Outputs
Clear	**Clock**	**D**	**Q**
L	X	X	L
H	↑	H	H
H	↑	L	L
H	L	X	Q_0

NOTES:
H = HIGH Level (Steady State)
L = LOW Level (Steady State)
X = Don't Care
↑ = Transition from LOW−to−HIGH level
Q_0 = The level of Q before the indicated steady state input conditions were established.

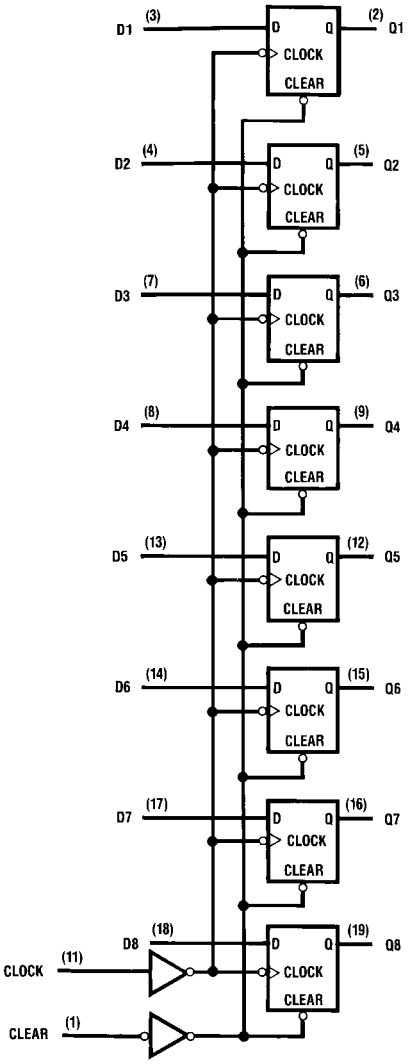


Figure 2. Logic Diagram

ABSOLUTE MAXIMUM RATINGS (Note 1)

Symbol	Rating		Value	Unit
V_{CC}	Supply Voltage		−0.5 to +6.5 V	V
V_{IN}	DC Input Voltage		−0.5 to V_{CC} +0.5 V	V
V_{OUT}	DC Output Voltage		−0.5 to V_{CC} +0.5 V	V
I_{IK}, I_{OK}	Clamp Diode Current		±20	mA
I_{OUT}	DC Output Current, per pin		±25	mA
I_{CC}	DC V_{CC} or GND Current, per pin		±50	mA
T_{STG}	Storage Temperature Range		−65 to +150	°C
P_D	Power Dissipation	SOIC	1302	mW
		TSSOP	833	mW
T_L	Lead Temperature (Soldering 10 seconds)		260	°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.
1. Unless otherwise specified all voltages are referenced to ground.

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RECOMMENDED OPERATING CONDITIONS

Symbol	Parameter		Min	Max	Unit
V_{CC}	Supply Voltage		2	6	V
V_{IN}, V_{OUT}	DC Input or Output Voltage		0	V_{CC}	V
T_A	Operating Temperature Range		−55	125	°C
t_r, t_f	Input Rise or Fall Times	V_{CC} = 2.0 V	–	1000	ns
		V_{CC} = 4.5 V	–	500	ns
		V_{CC} = 6.0 V	–	400	ns

Functional operation above the stresses listed in the Recommended Operating Ranges is not implied. Extended exposure to stresses beyond the Recommended Operating Ranges limits may affect device reliability.

DC ELECTRICAL CHARACTERISTICS (Note 2)

Symbol	Parameter	Conditions	V_{CC}	T_A = 25°C	T_A = −40 to 85°C	T_A = −55 to 125°C	Unit
				Typ	Guaranteed Limits		
V_{IH}	Minimum HIGH Level Input Voltage		2.0 V 4.5 V 6.0 V		1.5 3.15 4.2	1.5 3.15 4.2	1.5 3.15 4.2	V V V
V_{IL}	Maximum LOW Level Input Voltage		2.0 V 4.5 V 6.0 V		0.5 1.35 1.8	0.5 1.35 1.8	0.5 1.35 1.8	V V V
V_{OH}	Minimum HIGH Level Output Voltage	V_{IN} = V_{IH} or V_{IL} $\|I_{OUT}\| \le 20$ μA	2.0 V 4.5 V 6.0 V	2.0 4.5 6.0	1.9 4.4 5.9	1.9 4.4 5.9	1.9 4.4 5.9	V V V
		V_{IN} = V_{IH} or V_{IL} $\|I_{OUT}\| \le 4.0$ mA $\|I_{OUT}\| \le 5.2$ mA	 4.5 V 6.0 V	 4.2 5.7	 3.98 5.48	 3.84 5.34	 3.7 5.2	 V V
V_{OL}	Maximum LOW Level Output Voltage	V_{IN} = V_{IH} or V_{IL} $\|I_{OUT}\| \le 20$ μA	2.0 V 4.5 V 6.0 V	0 0 0	0.1 0.1 0.1	0.1 0.1 0.1	0.1 0.1 0.1	V V V
		V_{IN} = V_{IH} or V_{IL} $\|I_{OUT}\| \le 4.0$ mA $\|I_{OUT}\| \le 5.2$ mA	 4.5 V 6.0 V	 0.2 0.2	 0.26 0.26	 0.33 0.33	 0.4 0.4	 V V
I_{IN}	Maximum Input Current	V_{IN} = V_{CC} or GND	6.0 V		±0.1	±1.0	±1.0	μA
I_{CC}	Maximum Quiescent Supply Current	V_{IN} = V_{CC} or GND I_{OUT} = 0 μA	6.0 V		8	80	160	μA

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

2. For a power supply of 5 V ±10% the worst case output voltages (V_{OH}, and V_{OL}) occur for HC at 4.5 V. Thus the 4.5 V values should be used when designing with this supply. Worst case V_{IH} and V_{IL} occur at V_{CC} = 5.5 V and 4.5 V respectively. (The V_{IH} value at 5.5 V is 3.85 V.) The worst case leakage current (I_{IN}, I_{CC}, and I_{OZ}) occur for CMOS at the higher voltage and so the 6.0 V values should be used.

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AC ELECTRICAL CHARACTERISTICS

(V_{CC} = 5 V, T_A = 25°C, C_L = 15 pF, t_r = t_f = 6 ns)

Symbol	Parameter	Conditions	Typ	Guaranteed Limit	Unit
f_{MAX}	Maximum Operating Frequency		50	30	MHz
t_{PHL}, t_{PLH}	Maximum Propagation Delay, Clock to Output		18	27	ns
t_{PHL}	Maximum Propagation Delay, Clear to Output		18	27	ns
t_{REM}	Minimum Removal Time, Clear to Clock		10	20	ns
t_s	Minimum Setup Time, Data to Clock		10	20	ns
t_H	Minimum Hold Time, Clock to Data		−2	0	ns
t_W	Minimum Pulse Width, Clock to Clear		10	16	ns

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

AC ELECTRICAL CHARACTERISTICS

(C_L = 50 pF, t_r = t_f = 6 ns, unless otherwise specified)

Symbol	Parameter	Conditions	V_{CC}	T_A = 25°C	T_A = −40 to 85°C	T_A = −55 to 125°C	Unit
				Typ	Guaranteed Limits		
f_{MAX}	Maximum Operating Frequency		2.0 V 4.5 V 6.0 V	16 74 78	5 27 31	4 21 24	3 18 20	MHz MHz MHz
t_{PHL}, t_{PLH}	Maximum Propagation Delay, Clock to Output		2.0 V 4.5 V 6.0 V	38 14 12	135 27 23	170 34 29	205 41 35	ns ns ns
t_{PHL}	Maximum Propagation Delay, Clear to Output		2.0 V 4.5 V 6.0 V	42 19 18	135 27 23	170 34 29	205 41 35	ns ns ns
t_{REM}	Minimum Removal Time, Clear to Clock		2.0 V 4.5 V 6.0 V	0 0 0	25 5 4	32 6 5	37 7 6	ns ns ns
t_s	Minimum Setup Time, Data to Clock		2.0 V 4.5 V 6.0 V	26 7 5	100 20 17	125 25 21	150 30 25	ns ns ns
t_H	Minimum Hold Time, Clock to Clock		2.0 V 4.5 V 6.0 V	−15 −6 −4	0 0 0	0 0 0	0 0 0	ns ns ns
t_W	Minimum Pulse Width, Clock or Clear		2.0 V 4.5 V 6.0 V	34 11 10	80 16 14	100 20 18	120 24 20	ns ns ns
t_r, t_f	Maximum Input Rise and Fall Time, Clock		2.0 V 4.5 V 6.0 V	– – –	1000 500 400	1000 500 400	1000 500 400	ns ns ns
t_{THL}, t_{TLH}	Maximum Output Rise and Fall Time		2.0 V 4.5 V 6.0 V	28 11 9	75 15 13	95 19 16	110 22 19	ns ns ns

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AC ELECTRICAL CHARACTERISTICS (continued)

(C_L = 50 pF, $t_r = t_f$ = 6 ns, unless otherwise specified)

Symbol	Parameter	Conditions	V_{CC}	T_A = 25°C	T_A = −40 to 85°C		T_A = −55 to 125°C	Unit
				Typ	Guaranteed Limits			
C_{PD}	Power Dissipation Capacitance (Note 3)	(per flip−flop)		45				pF
C_{IN}	Maximum Input Capacitance			7	10	10	10	pF

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

3. C_{PD} determines the no load dynamic power consumption, $P_D = C_{PD} V_{CC}^2 f + I_{CC} V_{CC}$, and the no load dynamic current consumption, $I_S = C_{PD} V_{CC} f + I_{CC}$.

ORDERING INFORMATION

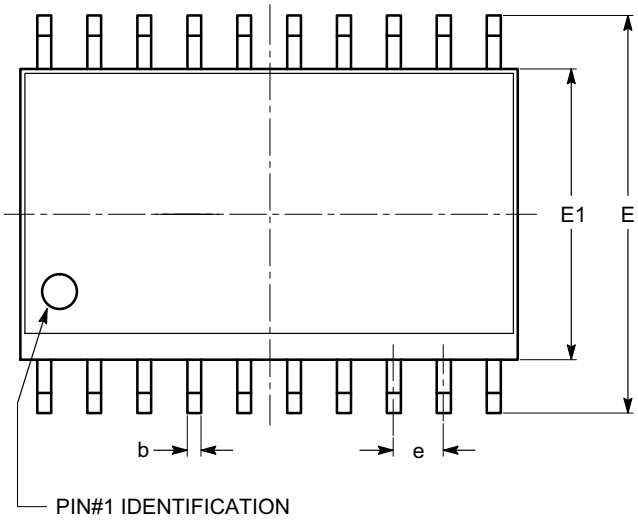
Device	Marking	Package	Shipping†
MM74HC273WM	HC273A	SOIC−20 WB (Pb−Free and Halide Free)	38 Units / Tube
MM74HC273WMX	HC273A	SOIC−20, 300 mils (Pb−Free and Halide Free)	1000 / Tape & Reel
MM74HC273MTC	HC 273A	TSSOP−20 WB (Pb−Free)	75 Units / Tube
MM74HC273MTCX	HC 273A	TSSOP−20 WB (Pb−Free)	2500 / Tape & Reel

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

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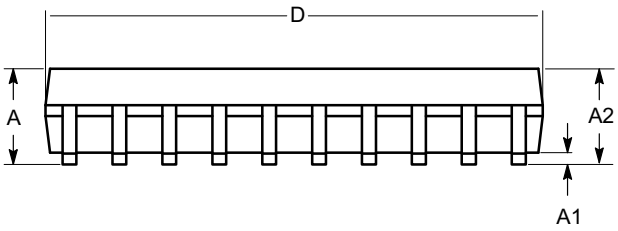
SOIC−20, 300 mils
CASE 751BJ
ISSUE O

DATE 19 DEC 2008

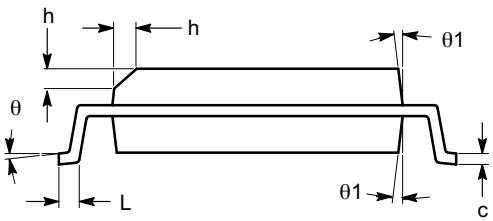


TOP VIEW

SYMBOL	MIN	NOM	MAX
A	2.36	2.49	2.64
A1	0.10		0.30
A2	2.05		2.55
b	0.31	0.41	0.51
c	0.20	0.27	0.33
D	12.60	12.80	13.00
E	10.01	10.30	10.64
E1	7.40	7.50	7.60
e		1.27 BSC	
h	0.25		0.75
L	0.40	0.81	1.27
θ	0°		8°
θ1	5°		15°



SIDE VIEW



END VIEW

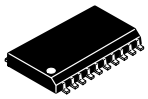
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(1) All dimensions are in millimeters. Angles in degrees.
(2) Complies with JEDEC MS-013.

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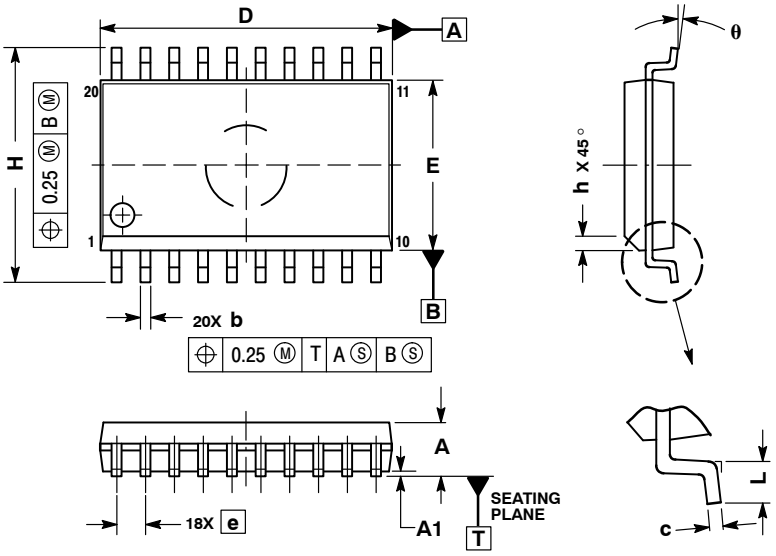
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SOIC−20 WB
CASE 751D−05
ISSUE H

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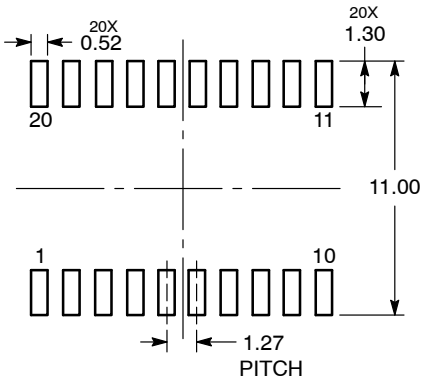
SCALE 1:1



NOTES:
1. DIMENSIONS ARE IN MILLIMETERS.
2. INTERPRET DIMENSIONS AND TOLERANCES PER ASME Y14.5M, 1994.
3. DIMENSIONS D AND E DO NOT INCLUDE MOLD PROTRUSION.
4. MAXIMUM MOLD PROTRUSION 0.15 PER SIDE.
5. DIMENSION B DOES NOT INCLUDE DAMBAR PROTRUSION. ALLOWABLE PROTRUSION SHALL BE 0.13 TOTAL IN EXCESS OF B DIMENSION AT MAXIMUM MATERIAL CONDITION.

	MILLIMETERS	
DIM	MIN	MAX
A	2.35	2.65
A1	0.10	0.25
b	0.35	0.49
c	0.23	0.32
D	12.65	12.95
E	7.40	7.60
e	1.27 BSC	
H	10.05	10.55
h	0.25	0.75
L	0.50	0.90
θ	0°	7°

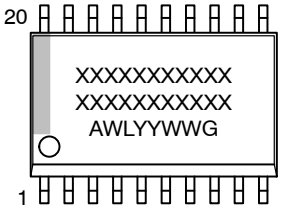
RECOMMENDED SOLDERING FOOTPRINT*



DIMENSIONS: MILLIMETERS

*For additional information on our Pb−Free strategy and soldering details, please download the **onsemi** Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

GENERIC MARKING DIAGRAM*



XXXXX = Specific Device Code
A = Assembly Location
WL = Wafer Lot
YY = Year
WW = Work Week
G = Pb−Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “▪”, may or may not be present. Some products may not follow the Generic Marking.

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